

Getting a Better Understanding of the System Log Architecture



System logs greatly aid sysadmins in keeping the system running like a well-oiled machine. Since an ounce of prevention is better than a pound of cure, monitoring system logs can help to anticipate problems and prevent breakdowns. One needs some knowledge of the system log architecture, however, to do this.

Processes and the operating system kernel need to be able to record a log of events that happen. These logs can be useful for auditing the system and for troubleshooting. By convention, the `/var/log` directory is where these logs are stored permanently.

A standard logging system based on the Syslog protocol is built into Red Hat Enterprise Linux (RHEL). Many programs use this system to record events and organise them into log files. In RHEL 7, syslog messages are handled by two services — *systemd-journald* and *rsyslog*.

The *systemd-journald* daemon provides an improved log management service that collects messages from the kernel, the early stages of the boot process, standard output and the error of daemons as they start up and run the syslog. It writes these messages to a structured journal of events that, by default, does not persist between reboots. This allows syslog messages and events that are missed out by the syslog to be collected in one central database. The syslog messages are also forwarded by *systemd-journald* to *rsyslog* for further processing.

The *rsyslog* service then sorts the syslog messages by type (or facility) and priority, and writes them to persistent files in the `/var/log` directory. The `/var/log` directory holds various system and service-specific log files maintained by *rsyslog*, as shown in Table 1.

Table 1: Overview of system log files

Log file	Purpose
<code>/var/log/messages</code>	Most syslog messages are logged here. The exceptions are messages related to authentication and email processing, which periodically run jobs, and those that are purely debugging related.
<code>/var/log/secure</code>	This is the log file for security and authentication related messages and errors.
<code>/var/log/maillog</code>	This is the log file with mail server related messages.
<code>/var/log/cron</code>	This is the log file related to periodically executed tasks.
<code>/var/log/boot.log</code>	Messages related to system startup are logged here.

Reviewing syslog files

Many programs use the Syslog protocol to log events of the system. Each log message is categorised by a facility (the type of message) and a priority (the severity of the message). The facilities that are available are documented by the *rsyslog.conf(5)* man page. The eight priorities are also standardised and ranked, as shown in Table 2.